

Core Pure Mathematics 2

Notes · Edexcel A Level Further Mathematics — Core Pure 2 · Chapter 2: Series

Chapter 2: Series

Why this matters

You already know how to differentiate standard functions and expand $(1 + x)^n$ for integer n using the binomial theorem. This chapter generalises both ideas: **Maclaurin series** let us write almost any "nice" function as an infinite polynomial built entirely from its derivatives at $x = 0$, and the **generalised binomial expansion** extends $(1 + x)^n$ to any rational n . These series are the backbone of approximation techniques used throughout further maths, physics, and numerical methods.

Concept: Maclaurin series — the idea

If a function $f(x)$ can be written as a power series and is infinitely differentiable at $x = 0$, then:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots + \frac{f^{(n)}(0)x^n}{n!} + \dots$$

This is the Taylor series of f centred at 0. The idea: match the value and every derivative of the polynomial to those of f at $x = 0$, so the polynomial "hugs" the curve of f as closely as possible near that point.

What this means in practice. Each extra term makes the polynomial a better copy of f near $x = 0$: the constant term matches the *height* $f(0)$, the linear term matches the *slope* $f'(0)$, the quadratic term matches the *curvature* $f''(0)$, and so on — each derivative pins down one more feature of the graph at that single point. That's why a Maclaurin polynomial is most accurate close to $x = 0$ and gets worse the further away you go — it's only guaranteed to match f and all its derivatives exactly *at* $x = 0$, not anywhere else. Truncating after a few terms (as every worked example below does) gives a polynomial **approximation**, useful for estimating values of $f(x)$ for small x without a calculator, or for simplifying a hard expression before integrating or taking a limit.

Key formula

KEY FORMULA

$$\text{Maclaurin series: } f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

Worked example 1 — Deriving the Maclaurin series for e^x

WORKED EXAMPLE 1

Question. Find the Maclaurin series for $f(x) = e^x$ up to and including the term in x^4 .

Step 1. Differentiate repeatedly: $f(x) = e^x$, $f'(x) = e^x$, $f''(x) = e^x$, $f'''(x) = e^x$, $f^{(4)}(x) = e^x$. **(M1)**

Step 2. Evaluate each at $x = 0$: $f(0) = f'(0) = f''(0) = f'''(0) = f^{(4)}(0) = 1$ (since $e^0 = 1$). **(A1)**

Step 3. Substitute into the Maclaurin formula: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ **(M1)(A1)**

Answer: $e^x = 1 + x + x^2/2 + x^3/6 + x^4/24 + \dots$ (valid for all x)

EXAM TIP

e^x is the natural first example because every derivative is identical, so you can focus entirely on the mechanism of the formula rather than getting bogged down in differentiation. Use it to learn the pattern $f^{(n)}(0)/n!$ before moving to functions where the derivatives actually change value and sign.

Worked example 2 — Deriving the Maclaurin series for $\sin x$

WORKED EXAMPLE 2

Question. Find the Maclaurin series for $f(x) = \sin x$ up to and including the term in x^5 .

Step 1. Differentiate and evaluate at $x = 0$, cycling through the pattern: **(M1)**

n	$f^{(n)}(x)$	$f^{(n)}(0)$
0	$\sin x$	0
1	$\cos x$	1
2	$-\sin x$	0
3	$-\cos x$	-1
4	$\sin x$	0
5	$\cos x$	1

(A1) — correct pattern of derivatives

Step 2. Substitute the nonzero values into the Maclaurin formula: **(M1)** $\sin x = 0 + 1 \cdot x + 0 - 1 \cdot x^3/3! + 0 + 1 \cdot x^5/5! - \dots$

Answer: $\sin x = x - x^3/6 + x^5/120 - \dots$ (valid for all x) **(A1)**

EXAM TIP

Notice immediately that only odd powers of x appear, because $\sin x$ is an odd function ($\sin(-x) = -\sin x$), while $\cos x$'s series will contain only even powers, since $\cos$ is even. This symmetry check is a fast way to sanity-check your own derivations in an exam, without redoing all the differentiation.

Key formula — standard Maclaurin series (learn these; you may be asked to derive or just quote them)

KEY FORMULA

Function	Series	Valid for
e^x	$1 + x + x^2/2! + x^3/3! + x^4/4! + \dots$	all x
$\sin x$	$x - x^3/3! + x^5/5! - \dots$	all x
$\cos x$	$1 - x^2/2! + x^4/4! - \dots$	all x
$\ln(1 + x)$	$x - x^2/2 + x^3/3 - x^4/4 + \dots$	$-1 < x \leq 1$
$(1 + x)^n$	$1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \dots$	$ x < 1$ (n not a positive integer)

Worked example 3 — Deriving the series for $\ln(1 + x)$

WORKED EXAMPLE 3

Question. Find the Maclaurin series for $f(x) = \ln(1 + x)$ up to the term in x^4 , and state the range of x for which it is valid.

Step 1. Differentiate: $f(x) = \ln(1+x)$, $f'(x) = (1+x)^{-1}$, $f''(x) = -(1+x)^{-2}$, $f'''(x) = 2(1+x)^{-3}$, $f''''(x) = -6(1+x)^{-4}$. **(M1)**

Step 2. Evaluate at $x = 0$: $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, $f'''(0) = 2$, $f''''(0) = -6$. **(A1)**

Step 3. Substitute into the formula, dividing by the correct factorials: **(M1)** $\ln(1+x) = 0 + x + (-1)x^2/2! + 2x^3/3! + (-6)x^4/4! + \dots = x - x^2/2 + x^3/3 - x^4/4 + \dots$ **(A1)**

Step 4. State the range of validity: the series converges for $-1 < x \leq 1$. **(B1) — this must be quoted, not derived from scratch in an exam**

Answer: $\ln(1 + x) = x - x^2/2 + x^3/3 - x^4/4 + \dots$, valid for $-1 < x \leq 1$.

Concept: the generalised binomial expansion

You've already met this formula for positive integer n at GCSE/A Level, where it terminates after $n+1$ terms (try $n = 2$ in the formula and you'll get the familiar $1 + 2x + x^2$). Here, for $(1 + x)^n$ where n is **not** a positive integer (e.g. negative, or a fraction), the expansion does not terminate:

$$(1 + x)^n = 1 + nx + n(n-1)x^2/2! + n(n-1)(n-2)x^3/3! + \dots, \text{ valid for } |x| < 1$$

This is itself the Maclaurin series of $(1+x)^n$ — you can derive it the same way as above, or simply quote and apply the standard result. For an expression like $(a + bx)^n$, always factor out a^n first so that you are expanding $(1 + (b/a)x)^n$, and adjust the range of validity accordingly ($|(b/a)x| < 1$).

Worked example 4 — Binomial (Maclaurin) expansion with a non-integer power

WORKED EXAMPLE 4

Question. Find the first four terms of the expansion of $1/\sqrt{4 - x}$, and state the range of x for

which the expansion is valid.

Step 1. Rewrite in the form $(1 + \dots)^n$: $1/\sqrt{4-x} = (4-x)^{-1/2} = 4^{-1/2}(1-x/4)^{-1/2} = \frac{1}{2}(1-x/4)^{-1/2}$. **(M1) — factoring out 4 correctly**

Step 2. Apply the standard expansion with $n = -1/2$ and x replaced by $(-x/4)$: **(M1)** $(1-x/4)^{-1/2} = 1 + (-1/2)(-x/4) + (-1/2)(-3/2)(-x/4)^2/2! + (-1/2)(-3/2)(-5/2)(-x/4)^3/3! + \dots$

Step 3. Simplify each term: **(A1)(A1) — correct simplification of coefficients** $= 1 + x/8 + (3/8)(x^2/16) + (-5/16)(-x^3/64) + \dots = 1 + x/8 + 3x^2/128 + 5x^3/1024 + \dots$

Step 4. Multiply back by the $\frac{1}{2}$ factored out in Step 1: **(M1)** $\frac{1}{2}(1 + x/8 + 3x^2/128 + 5x^3/1024 + \dots) = \frac{1}{2} + x/16 + 3x^2/256 + 5x^3/2048 + \dots$

Step 5. State validity: need $|x/4| < 1$, i.e. **(B1)**

Answer: $1/\sqrt{4-x} \approx \frac{1}{2} + x/16 + 3x^2/256 + 5x^3/2048 + \dots$, valid for $-4 < x < 4$.

EXAM TIP

The single biggest source of lost marks here is forgetting to adjust the validity range after factoring — quoting “ $|x| < 1$ ” from memory instead of $|x/4| < 1$. Always write the substituted variable explicitly (here, $u = -x/4$) and solve $|u| < 1$ for x , rather than trying to remember a rule.

Concept: combining series (products, substitutions, and composite functions)

Once you know the standard series, you can build new ones by substitution (e.g. replacing x with x^2 in $\cos x$'s series to get the series for $\cos(x^2)$) or by multiplying two series together and collecting terms up to the required power. Always be careful to keep only the terms up to the power the question asks for — a correct method with an early term dropped will still lose accuracy marks.

Substitution also changes the range of validity, and this is easy to miss. If you substitute $x \rightarrow x^2$ into a series valid for *all* x (like $\sin x$ or e^x), the result is still valid for all x . But if you substitute into a series with a restricted range — e.g. replacing x with x^2 in $\ln(1+x)$ (valid $-1 < x \leq 1$) to get the series for $\ln(1+x^2)$ — you need $-1 < x^2 \leq 1$, which gives $-1 \leq x \leq 1$, not the original range unchanged. Always re-solve the inequality in terms of the new variable rather than copying the old range across.

Worked example 5 — Combining two Maclaurin series

WORKED EXAMPLE 5

Question. Using the standard series for e^x and $\ln(1+x)$, find the Maclaurin expansion of $e^x \ln(1+x)$ up to and including the term in x^3 .

Step 1. Write down each series up to x^3 : **(B1)** $e^x \approx 1 + x + x^2/2 + x^3/6$ $\ln(1+x) \approx x - x^2/2 + x^3/3$

Step 2. Multiply the two series, keeping only terms up to x^3 : **(M1)** $(1 + x + x^2/2 + x^3/6)(x - x^2/2 + x^3/3)$

Step 3. Collect contributions to each power systematically: **(M1)(A1) — correct collection of x^1 , x^2 , x^3 terms**

- x^1 term: $1 \cdot x = x$
- x^2 term: $1 \cdot (-x^2/2) + x \cdot x = -x^2/2 + x^2 = x^2/2$
- x^3 term: $1 \cdot (x^3/3) + x \cdot (-x^2/2) + (x^2/2) \cdot x = x^3/3 - x^3/2 + x^3/2 = x^3/3$

Answer: $e^x \ln(1+x) = x + x^2/2 + x^3/3 + \dots$ (up to the term in x^3) **(A1)**

Chapter summary

Idea	Key point
Maclaurin series	$f(x) = f(0) + f'(0)x + f''(0)x^2/2! + \dots$ — a Taylor series centred at $x = 0$
Standard series to memorise	e^x , $\sin x$, $\cos x$, $\ln(1+x)$, $(1+x)^n$ — see table above
Generalised binomial	$(1+x)^n$ for non-integer n ; only valid for $ x < 1$; must factor for $(a+bx)^n$
Combining series	Substitute into known series, or multiply/add series term-by-term up to the required power

Common exam mistakes

- Mixing up $n!$ in the denominator — writing x^n/n instead of $x^n/n!$ (or vice versa) in the general Maclaurin formula.
- Forgetting to state the range of validity for $\ln(1+x)$ and $(1+x)^n$ expansions — this is often a separate mark.
- Not adjusting the range of validity after substituting or factoring (e.g. for $(1+2x)^n$ or $(4-x)^n$).
- Assuming a substituted series (e.g. $x \rightarrow x^2$) keeps the same range of validity as the original — the inequality must be re-solved for the new variable.
- When multiplying two series, including terms beyond the required power (wasted work) or dropping a contributing term below it (lost accuracy).
- Sign errors in the alternating series ($\sin x$, $\cos x$, $\ln(1+x)$) — a clear derivative table avoids this.

You should now be able to

- State and derive the Maclaurin series for e^x , $\sin x$, $\cos x$, and $\ln(1+x)$ from first principles.
- Use the generalised binomial expansion for $(1+x)^n$ with non-integer n , including stating the range of validity.
- Adapt standard series by substitution to find the series of related functions.
- Multiply or combine known series to find the expansion of a product or composite function up to a given power.